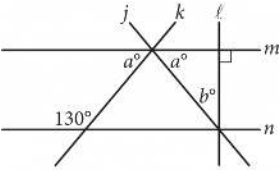


Question ID: 3828f53d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.
In the figure above, lines m and n are parallel. What is the value of b ?

Answer

- A. 40
- B. 50
- C. 65
- D. 80

Correct Answer: A

Rationale

Choice A is correct. Given that lines m and n are parallel, the angle marked 130° must be supplementary to the leftmost angle marked a° because they are same-side interior angles. Therefore, $130^\circ + a^\circ = 180^\circ$, which yields $a = 50^\circ$. Lines j and l intersect line m at a right angle, so lines j, l, and m form a right triangle where the two acute angles are a° and b° . The acute angles of a right triangle are complementary, so $a^\circ + b^\circ = 90^\circ$, which yields $50^\circ + b^\circ = 90^\circ$, and $b = 40$.

Choice B is incorrect. This is the value of a, not b. Choice C is incorrect and may be the result of dividing 130° by 2. Choice D is incorrect and may be the result of multiplying b by 2.